JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

An Autonomous Institute, with NAAC "A" Grade

Basic Science & Humanities Department

2023-24 (EVEN Semester)

Education to Eternity

MSE EXAMINATION

Course: B. Tech.
Subject: Data Structure
Subject Code: CS2T006
Year/Semester: IInd Semester (First Year)
Max Marks: 20 **Date : 24/04/2024** **Duration:- 1 Hr**

Instructions to the Students:
1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

CO1.To understand basic data structures, their implementation and some of their standard applications.
CO2. To describe basic data structures as arrays and array operation.
CO3. To analyze the types of data structures and its operations.
CO4. To apply linear data structures to various computing problems.
CO5. To develop programs for Sorting and Searching.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Arrays	1/CO1	1
Q.2	State Time Complexity	1/CO1	1
Q.3	Give the example of Non-Linear Data Structure.	2/CO1	1
Q.4	State the definition of Stack.	1/CO1	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Discuss the various Data Structure Operations.	2/CO2	2
Q.6	Express representation of an array with example.	2/CO2	2
Q.7	Write short notes on Space Complexity	1/CO2	2
Q.8	Illustrate the types of Non-Primitive Data Structure	3/CO1	2

SECTION C

1. **Solve any two questions.**

Q.9	Explain types of Data Structure.	2/CO1	4
Q.10	Write operation on Arrays in Data Structure	3/CO2	4
Q.11	Explain types of Arrays.	2/CO2	4